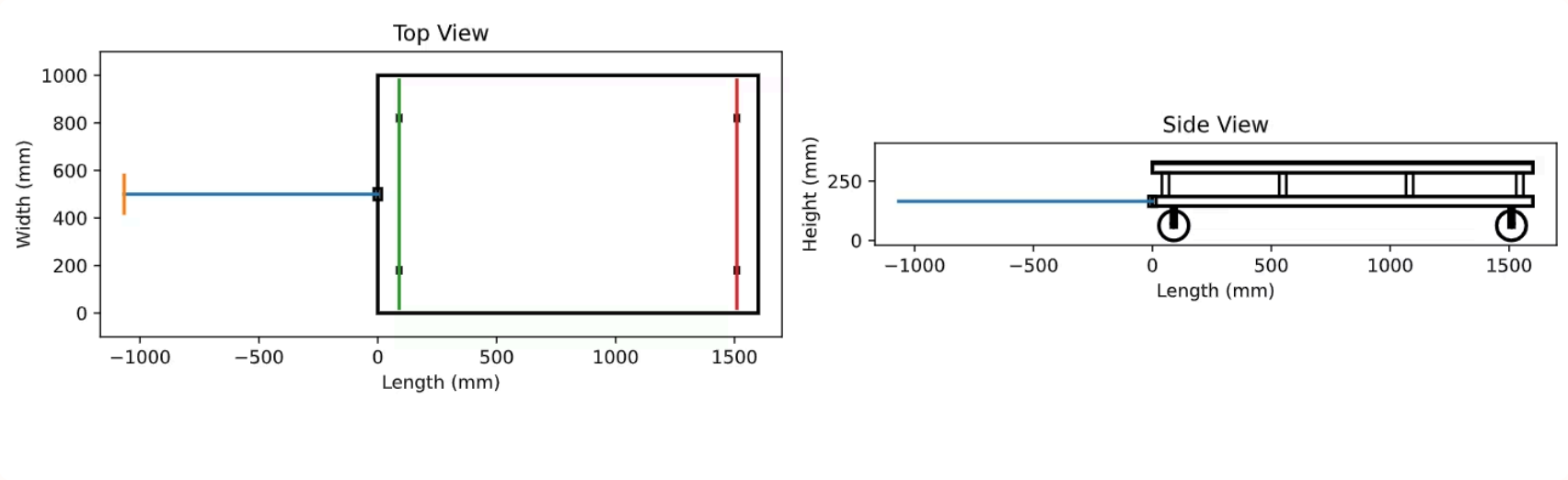
Material Handling Trolley – Engineering Calculation (physics based)



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1. Input Parameters

The following client requirements were used as input:

Load Capacity

900 kg

Platform Length

1600 mm

Platform Width

1000 mm

Material

Stainless Steel

Number of Wheels

4

Operating Environment

Indoor

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2. Key Outputs from Calculations

2.1 Design Parameters

Recommended Frame Tube: **60 × 60 × 4 mm**

Platform Thickness: **4.5 mm**

2.2 Mass Breakdown

- Frame Mass
- Deck (Plate) Mass
- Axle Mass
- Wheel Mass
- Payload: 900 kg
- Total Loaded Mass: Sum of all components

2.3 Mechanical Performance

- Load per Wheel (N)
- Rolling Resistance Force (N)

2.4 Stability Parameters

- Center of Gravity Height (mm)
- Tipping Angle (degrees)

2.5 Strength Parameters

- Yield Strength: **215 MPa**
- Safety Factor: **2.5**
- Allowable Stress: **~86 MPa**

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3. Physics Behind the Calculations

3.1 Material Properties

Mass is calculated using:

> **Mass = Volume × Density**

Material density is used to determine weight, and yield strength is used for safety calculations.

3.2 Frame Mass Calculation

The frame is modeled as hollow square tubes.

> Cross-sectional Area = Outer Area − Inner Area
>
> Total Tube Length = Perimeter + Cross Members
>
> Frame Mass = Density × Area × Length

3.3 Deck (Plate) Mass

The platform is treated as a flat rectangular plate:

> Deck Mass = Length × Width × Thickness × Density

3.4 Axle Mass

Axles are modeled as cylindrical rods:

> Area = πr^2
>
> Axle Mass = Density × Area × Length × Number of Axles

3.5 Total System Mass

> **Total Loaded Mass = Frame + Deck + Axles + Wheels + Payload**

This represents the actual operating weight of the trolley.

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3.6 – 3.9 Load, Resistance & Stability

3.6 Load Distribution

Assuming uniform load distribution:

> Load per Wheel = (Total Mass × g) / Number of Wheels

This ensures each wheel carries equal load under static conditions.

3.7 Rolling Resistance

The force required to move the trolley is:

> Rolling Force = μ × Normal Force

Where μ is the rolling resistance coefficient (**0.015** for indoor conditions).

3.8 Center of Gravity (CG)

The CG is calculated using a weighted average:

> CG = (Σ mass × height) / Total Mass

This determines the stability of the system.

3.9 Tipping Stability

Tipping occurs when CG crosses the wheel base edge:

> Tipping Angle = arctan((Base Width / 2) / CG Height)

A higher angle indicates better stability.

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3.10 Strength and Safety & Assumptions

3.10 Strength and Safety

Allowable stress is calculated as:

Allowable Stress = Yield Strength / Safety Factor

This ensures the structure operates within safe limits.

4. Assumptions Made

1. Load is uniformly distributed
2. Static loading conditions (no dynamic effects)
3. Material properties are standard
4. Equal load on all wheels
5. Simplified geometric modeling

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5. Limitations of the Model

The following limitations apply to this physics-based engineering calculation model:

5.1 Structural Limitations

- No beam bending or stress analysis
- No Finite Element Analysis (FEA)

5.2 Load Assumptions

- Ignores dynamic loads and impact forces
- Assumes perfect load distribution

5.3 Component Simplifications

- Vertical supports not explicitly included
- Fasteners and weld mass ignored

5.4 Wheel Analysis

- No bearing or wheel stress analysis

5.5 Manufacturing Considerations

- Fabrication tolerances not considered

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